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Project Description:

I built a render of a whale fall on the seafloor in three different and distinct visual styles. The first one is inspired from pictures from the movie, “Iron Lung”. The second is what it would look like if the whale skeleton was in shallower waters closer to the sun, and the third is where whale falls are usually located, in the deep sea.

Techniques Used:

- **Post Processing Shaders**: CRT Scanlines, Film Grain, Vignette, Dithering, Grey-Scale, Contrast
- **Procedural Generation**: Voronoi Cellular Noise, L-Systems
- Shell Texturing for fur shading
- Point Lighting

Steps:

I implemented the “Iron Lung”-inspired style first because that largely consisted of post-processing effects. Those were the easiest to implement, and it helped that our lab at the time was the one about post-processing.

Then, I worked on the shallow water mode. I mainly worked on mimicking water caustics because that was the main focal point of that style. When I was doing research on implementing real-time and realistic water caustics, I realized that it would be super hard, but I found a video on Youtube that used cellular noise to mimic water caustics instead. I chose that route and modified the Book of Shader’s tutorial on Voronoi Cellular Noise. Afterwards, since I had the time, I implemented L-Systems to procedurally generate plants/coral which was one of my stretch goals.

Finally, I implemented point lighting for the abyssal mode (which I then re-used for the Iron Lung mode). Another one of my stretch goals was implementing a fur shader to mimic the worms and deep-sea bacteria that grow on the bones. I adapted a tutorial for shell texturing, and that became my added challenge for what was originally an extremely simple render.

Challenges:

For most of the graphics techniques I used, I based my code on someone else's, but the hardest part was adapting it to look like how I wanted it to look like. This required a lot of experimental tweaking and understanding how the code works, so that I could modify it closer to what I wanted.

For example, the original code for the shell texturing uses a plane as the base geometry, but in my project, my base geometry is a sphere. As a result, there are certain angles where it doesn't render correctly, and I am still not sure why. One fix I tried was keeping the step of extruding along the surface's vertex normals, but after that, converting those extruded coordinates from Cartesian coordinates to Spherical coordinates. That didn't look much better than the original, but it is a step closer to figuring out how to orient the "hairs" the right way.

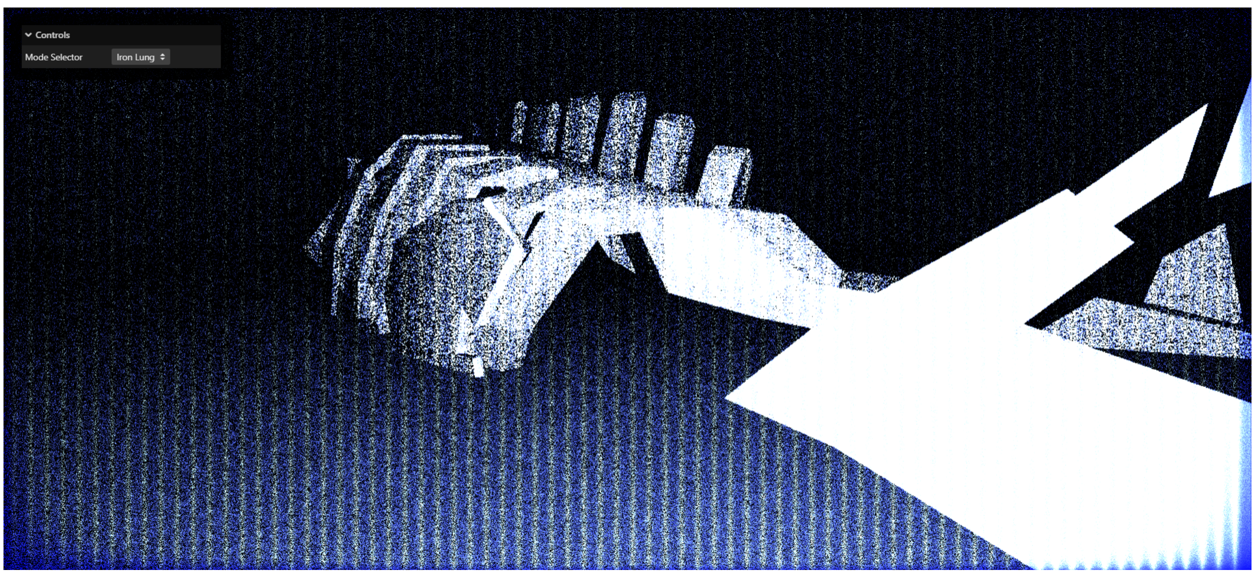
Another example is when I adapted the Voronoi Cellular Noise. The tutorial that Book of Shaders has looks nothing like water caustics at all, and this was the hardest effect to implement. One change I remember making is changing up how the uv coordinates are calculated. The original code for the cellular noise did not look like it was mapped to the floor, so by making a simple change, I was able to add perspective, therefore making it look more correct.

Another small example is changing the color of the CRT scanline effect. The original had the CRT scanlines as black because the way it's calculated uses multiplication, therefore deepening the pixel color. After some thought, I realized that in order to get white scanlines which are closer to my reference images, I had to divide the color instead to lighten it up.

One of the biggest challenges was keeping my codebase organized and clean. Every visual effect you can see is implemented solely with GLSL shaders, so with multiple objects in a scene where similar lighting needed to be applied to them, it became messy very quickly. One of my solutions was keeping track of global uniforms and injecting strings into the shaders to cut down on the amount of repeated code between shaders. This somewhat helped, but it is still very messy.

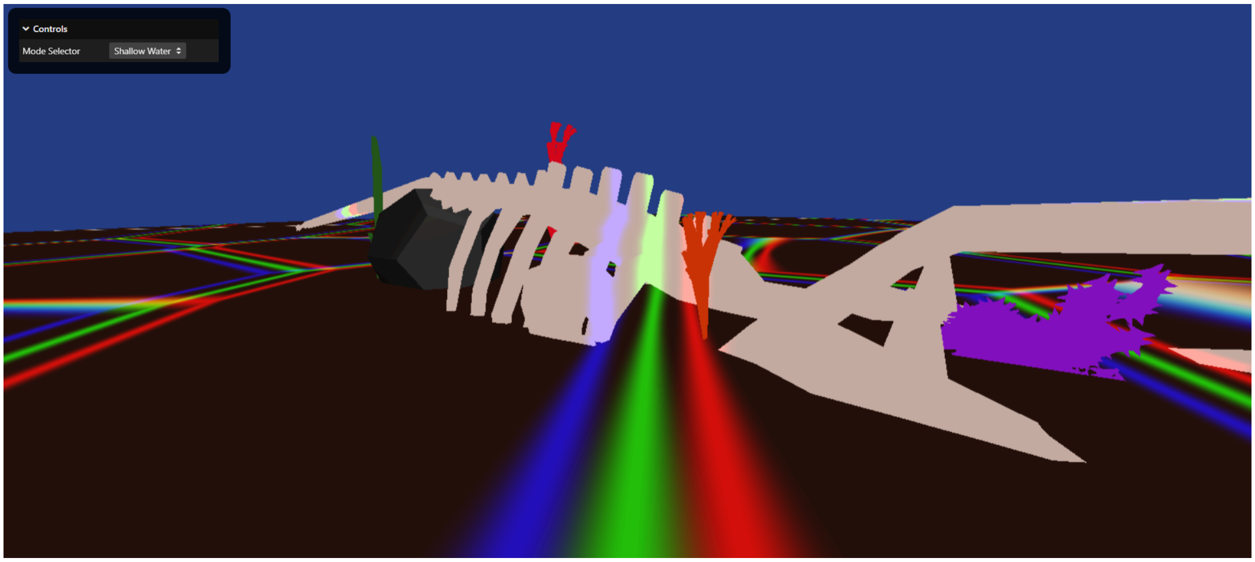
Results:

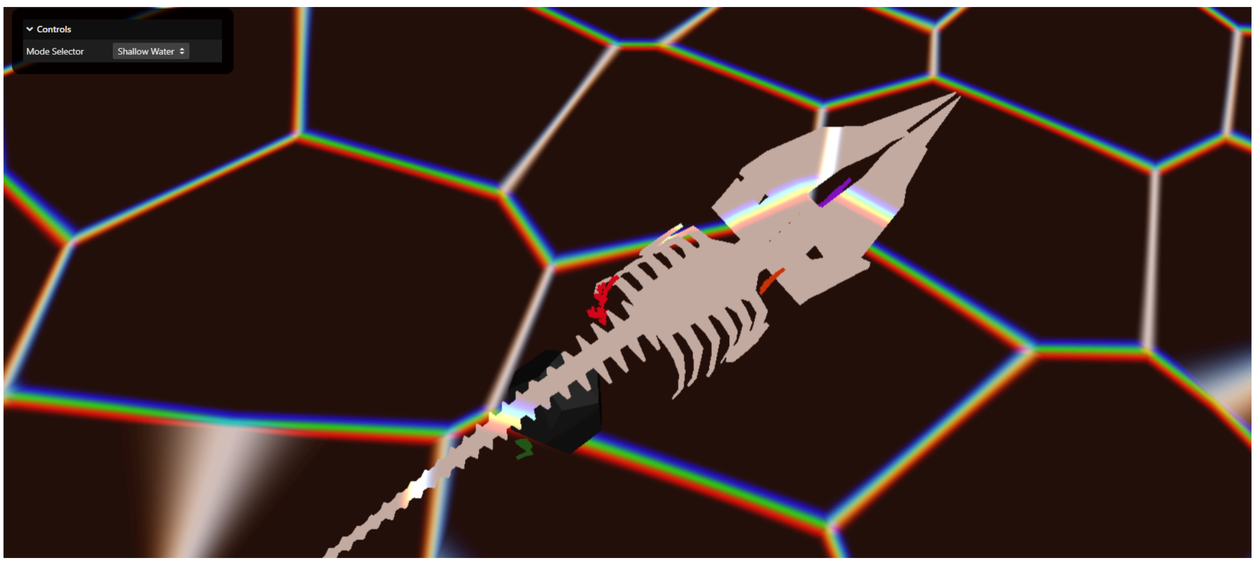
Reference



“Iron Lung” mode is my personal favorite, and I think I was able to somewhat accurately mimic the style of my reference image. There is film grain, white scanlines, the bright and highly contrasted lighting, and dithering as well.

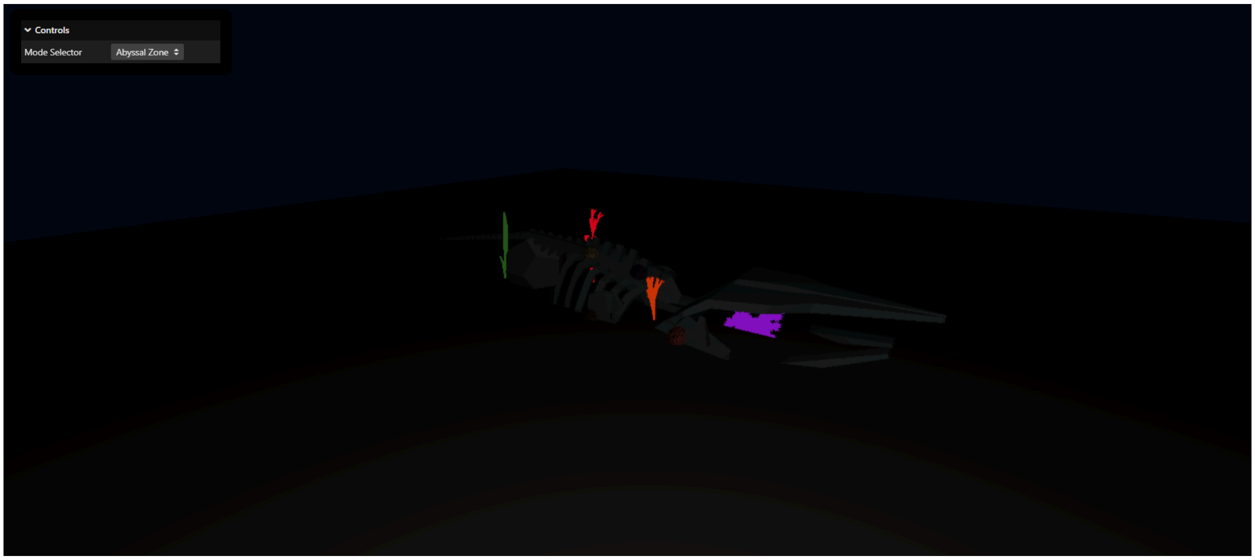
One thing I would change is making the edges of each object in this scene more jagged. They are too clean and crisp compared to the reference, so it doesn’t look accurate.

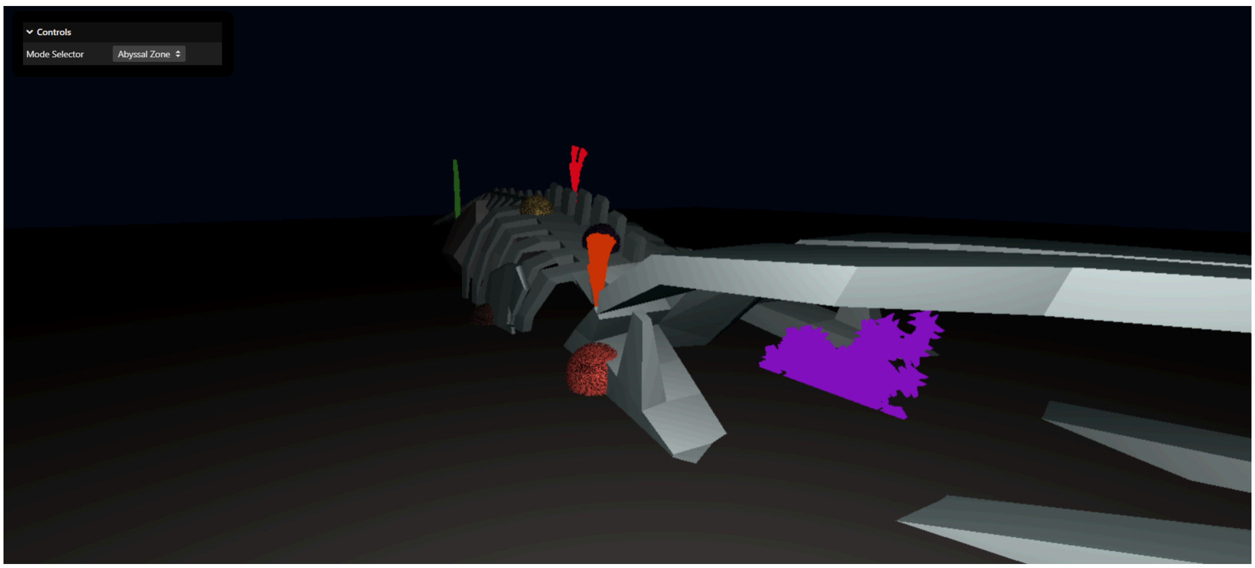


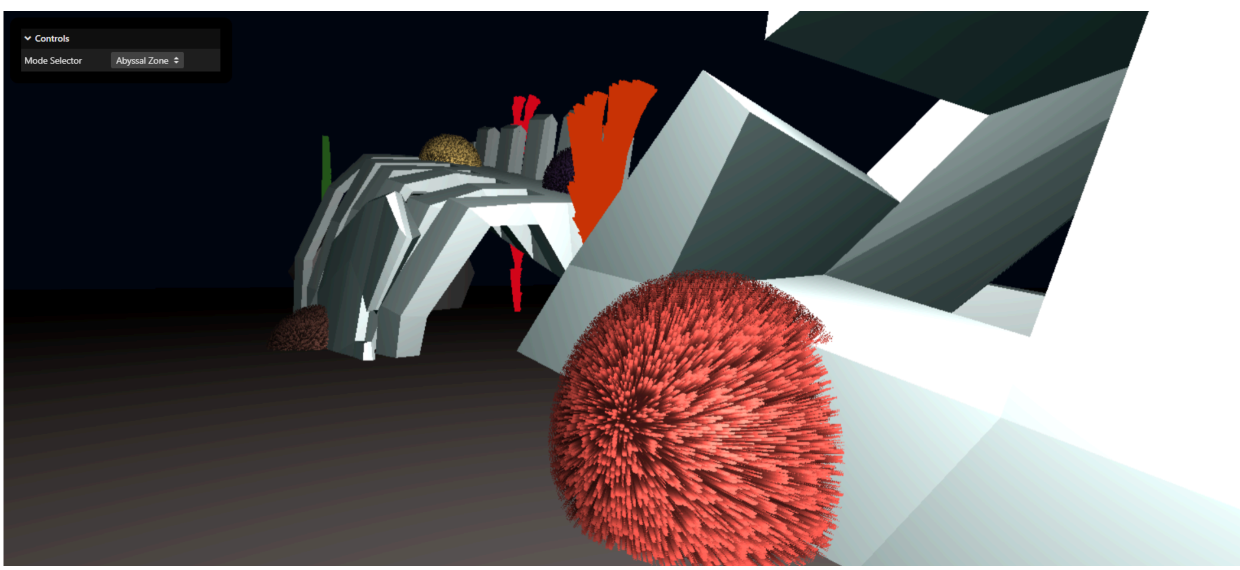


This one is my least favorite out of the three because I wasn't able to fully implement all the light through water effects I wanted. However, I was able to get the caustics looking somewhat close to real water caustics using techniques like layering and adding chromatic aberration.

I also really liked the plants I created using L-Systems. The red and orange plants are taken from the examples in the tutorial I used because I liked how they looked, but the purple and green plants were ones generated using a string that I wrote. The biggest thing I'd like to change is implementing realistic light in water effects like true water caustics and light rays shining through water. A simpler change I would like to make is adding proper lighting to the rock and the plants.


Controls
Mode Selector
Abyssal Zone


Controls
Mode Selector
Abyssal Zone


Controls
Mode Selector
Abyssal Zone

The point lighting shader works very well in this case, and it truly looks like you are in a submarine at the bottom of the sea. The fur shader also looks like fur, however, once you look at it from certain angles, it doesn't properly render.

That is the biggest change I want to make, figuring out how to make the fur render properly from every angle. Another change I want to make is figuring out how to implement a cloudy/foggy effect so that it looks like light is being reflected off of particles in the surrounding water.

Effort Justification:

During the course of the project, I learned how to implement many different graphics techniques, about ten in total, all from different categories, and not only that, I have also added my own modifications to many of them. Many of these techniques were also combined to create the effects I wanted.

For example, the tutorial for the shell texturing uses a modification of Valve's Half Lambert alongside faked ambient occlusion. For mine, I combined the ambient occlusion with the same point lighting shader used for the sea floor and whale skeleton so that the balls of fur also fade away into the darkness when the camera zooms out.

I also implemented everything through GLSL shaders as well. All the lighting is done through a shader, not through three.js's pre-built lights. Each of the post-processing effects was created through shaders as well, and while we did have a lab on post-processing which I did take some inspiration from, post-processing was done using a multi-pass render rather than through p5.js.

Rubric:

Points	Criteria
25	Implement various lighting and post-processing effects without using any built-in three.js effects
15	Lighting should look similar or very close to the reference images for each lighting mode
15	Implement interesting lighting effects like water caustics and combine several post-processing effects
15	Implement OrbitControls and controls like sliders that can be used to alter the way the scene looks
70	**Total (your rubric)**

25 + 12 + 15 + 8 = 60

25: I was able to implement everything through GLSL shaders and without using any built-in effects or lights or anything.

12: Lighting looks similar or very close in the Iron Lung and Abyssal styles, but in the style mimicking shallow waters, it is not close at all. However, the cellular noise at least looks like it is clearly mimicking water caustics.

15: I combined several post-processing effects and was able to implement lighting effects that we didn't learn in class like point lights.

8: I did add some interactivity through OrbitControls, but outside of switching between modes, there isn't much interactivity because I ran out of time and also wasn't able to come up with any cool ideas for interactivity outside of tweaking shader parameters.

Bibliography:

- CRT Scanline (Retro CRT Shader)
- Film Grain (Film Grain Shader)
- Voronoi Cellular Noise (More noise)
- Making Water Caustics in Reality Composer Pro Shader Graph (Beginner Friendly)
- L-Systems (Procedural Content Generation: L-Systems)
- Point Lighting (Light casters)
- Shell Texturing for Fur Shading (How Do Games Render Fur?)
- Whale Skeleton Model (Low Poly Whale Bones)